

An Investigation on Location in the Round of 64 in March Madness

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May 2026

1 Introduction

Every year, analysts, media personnel, and the fans of NCAA Men's College Basketball aim to predict the first round of the NCAA Men's Collegiate Basketball Championship, AKA March Madness. People consider many factors when predicting these results, one of which is location. If a team is closer to the location of their game, it seems logical that they might have a better chance of winning the game. This is typically seen with "upset picks," where if the inferior team is closer to the site of the game, they are touted as underdogs to watch or to bet on. This investigation takes a deep dive into March Madness data from 2011-2019 and 2022-2026 to see: **Does location give you an edge over the spread when betting?**

2 Key Terms to Know

To summarize the key terms:

- **Spread** - The amount of points the favorite is expected to win by.
 - Example 1: (-3.5) means a team is favored by 3.5 points. If they win by more than 3.5, the team "covers" and is seen as performing above expectations. If they win by less than 3.5 or lose outright, they "don't cover" and are seen as performing below expectations.
 - Example 2: (+6) means a team is an underdog by 6 points. If they lose by less than 6 or win outright, the team "covers" and is seen as performing above expectations. If they lose by exactly 6 points, it is considered a "push" and is seen as performing at expectations. If they lose by more than 6 points, they "don't cover" and are seen as performing below expectations.
- **Distance** - The straight line distance between the location of the game and the college of the team participating in the match (in miles).

- **Seed** - The "ranking" of each team in a given March Madness tournament. Out of the 64 teams, there are 4 groups of 16 teams seeded 1-16, with the 1 seed being the best team in the group, and the 16 seed being the worst. Matchups in the Round of 64 will always be a 1 v 16, 2 v 15, 3 v 14, ... 8 v 9.
- **First Four** - The first 4 games played in the tournament are to whittle the initial 68 team field to 64. These are usually the matchups to determine the 11 or 16 seeds. They are NOT apart of the Round of 64 but are apart of the tournament and are typically played a few days prior to the Thursday the Round of 64 starts.

3 Data Collection

3.1 Scores and Spread Data

The sports data needed for this were the scores of the games of the Round of 64 and the closing spread for the games. This was obtained through searches of score databases and various news articles covering the spreads at the time. Pre-2011 spread data could not be obtained and therefore is the reason why this investigation only goes to 2011.

It's to be noted that this is not a complete set of Round of 64 games from 2011-2026, as some games were purposefully and not purposefully excluded from this data. The 2021 games were not included because, due to the COVID-19 Pandemic, there were limited fans in the arena, largely only people with direct relations to the team. A large part of why a location advantage is important and not just a random factor with no contextual plausibility to dictate the outcome of the game is because of the fans. If one team has more fans than the other, the support from these fans improves player's mindsets and makes it closer to a home game environment. 2021 doesn't have this element involved so this study does not take these games into account.

Another set of games intentionally excluded are pushes. Pushes as described above are games in which no team covers the spread. If the favorite has a spread of (-6), and they win by 6, it's a push. Now, if every game had the possibility of being a push, they would be a legitimate 3rd outcome besides "cover" and "don't cover". However, most spreads are binary outcomes, cover vs doesn't cover, which occurs when the spread ends in .5. As such, there is no possibility for a push, and since every game can't end in a push, this study does not include games in which a push happened. Note that games in which a push was possible but didn't happen are here because there would have been the same binary outcome regardless.

The unintentionally excluded games are games for which no data was found. These primarily include 1 vs 16 and 6 vs 11 games in the earlier years because those matchups were only determined after the First Four games were played, meaning fewer articles were made with the spreads, and fewer records were kept.

The articles used from those years primarily cover the 28 games that had both teams determined when the 68-team field was announced.

3.2 Distance/Location Data

Now, when looking at distance, this study decided to use straight-line distance from the college to the city in which the game is held. The reason for this is that fans may travel to the games in various ways: plane, car, bus, etc. It is not possible to track how every fan is to travel to the game and average it out. Therefore, straight-line distance was used to eliminate that variation. This data was collected using Google Maps and Airline Travel Data. Distance is measured in miles.

4 Hypotheses

This study looks at three methods of seeing if location is a factor. First, it looks at all games in which one team had to travel more than double the distance of the other. In the attached Python file with the code that gave me the results, the metric used to determine this is called *DistRatio*. DistRatio was equal to the Distance the Lower Seed had to travel divided by the Distance the Higher Seed had to travel. So, the games that fit the double distance traveled requirement are any game with DistRatio greater than 2 or less than .5. Spreads are set to be where half the time a team would cover that line, and half the time a team would not. So, if location data were to give you an edge on spread betting, you could win over half the time. This allows us to generate the first set of hypotheses:

- **Null Hypothesis** : If a team has to travel less than half the distance of their opponent, they will cover the spread half the time.
- **Alternative Hypothesis** : If a team has to travel less than half the distance of their opponent, they will cover the spread MORE than half the time.

The second method, instead of looking at the multiplicative difference, looks at the additive difference between the two teams' distance traveled. Instead of picking an arbitrary distance cut-off, such as 300, 500, 750, or 1000 miles, across all 438 games used, the standard deviation of all the distance differences was used to determine the cutoff(619.88 miles). This leads to the second set of hypotheses:

- **Null Hypothesis** : If a team has to travel at least 619.88 miles less than their opponent, they will cover the spread half the time.
- **Alternative Hypothesis** : If a team has to travel at least 619.88 miles less than their opponent, they will cover the spread MORE than half the time.

The third method combines the previous two methods; both conditions must be met. A team must travel less than half the distance of their opponent, and the difference must be greater than 619.88 miles. The hypotheses for this test are as follows:

- **Null Hypothesis** : If a team travels less than half the distance of their opponent AND the difference is greater than 619.88 miles, they will cover the spread half the time.
- **Alternative Hypothesis** : If a team travels less than half the distance of their opponent AND the difference is greater than 619.88 miles, they will cover the spread MORE than half the time.

These are the 3 tests that the study will conduct, and it will conduct them on 7 groups. First, the whole sample will be evaluated, and all games that fit the criterion will be considered. Then, the sample will split into when the Low Seed has the location advantage and when the High Seed has the location advantage. Lastly, the study analyzes the 4 seed groups (1-4, 5-8, 9-12, 13-16).

5 Testing and Results

For each of the above hypotheses, the 7 groups were each evaluated with a one-sided binomial test. In total, 438 games were collected. The data was analyzed in Pandas, and using the stats module from Scipy, the tests were conducted. Here are the results:

Table 1: **Distance Ratio Analysis Results.** n = sample size of games that fit the criteria, x = number of games in which the team with the travel advantage covered the spread, Cover Rate = $100 \cdot x/n$, P-Value = result of one-sided binomial test against $p = 0.5$.

Group	n	x	Cover Rate (%)	P-Value
Whole	208	115	55.288	0.0726
High Seed	68	42	61.765	0.0341
Low Seed	140	73	52.143	0.3364
1-4	91	49	53.846	0.2648
5-8	49	24	48.980	0.5000
9-12	50	29	58.000	0.3222
13-16	18	13	72.222	0.0481

Table 2: **Standard Deviation Distance Analysis Results.** n = sample size of games that fit the criteria, x = number of games in which the team with the travel advantage covered the spread, Cover Rate = $100 \cdot x/n$, P-Value = result of one-sided binomial test against $p = 0.5$.

Group	n	x	Cover Rate (%)	P-Value
Whole	144	76	52.778	0.2799
High Seed	33	56	58.9286	0.1144
Low Seed	88	43	48.8636	0.4576
1-4	53	27	50.943	0.5
5-8	35	16	45.714	0.3679
9-12	42	25	59.524	0.1400
13-16	14	8	57.143	0.3953

Table 3: **Combined DistRatio and Standard Deviation Distance Analysis** n = sample size of games that fit the criteria, x = number of games in which the team with the travel advantage covered the spread, Cover Rate = $100 \cdot x/n$, P-Value = result of one-sided binomial test against $p = 0.5$.

Group	n	x	Cover Rate (%)	P-Value
Whole	118	64	54.237	0.2037
High Seed	44	26	59.091	0.1456
Low Seed	74	38	51.351	0.4538
1-4	47	24	51.064	0.5000
5-8	27	14	51.852	0.5000
9-12	33	19	57.576	0.2434
13-16	11	7	63.636	0.2744

6 Analysis of Results

All results will be interpreted with a significance level of 0.05

Most tests, including all the tests on the whole dataset, do NOT have $p < 0.05$. Therefore, as a whole, this investigation CAN'T make the general claim that location analysis gives a spread bettor an advantage/edge in the Round of 64 of the Men's March Madness Tournament. Moreover, all tests in the Standard Deviation Analysis did not have $p < 0.05$, meaning there is no evidence to say that additive location differences are something that bettors can use to give them an edge.

Now, the multiplicative difference tests are where certain advantages can be drawn from. High-seeded teams(9-16) have a 61.765 percent win rate(42/68) with $p = 0.0341$, meaning that there is evidence to suggest that in games where the higher-seeded team has a multiplicative location advantage, spread bettors have an edge. However, further analysis shows that this advantage primarily comes from 13-16 seeds. So, the investigation concludes that if a higher seed

has a multiplicative location advantage, if it's a 13-16 seed to definitely consider going for the spread bet, but if it's a 9-12 seed, proceed with caution and look at other factors as well.

None of the combined multiplicative and additive distance difference tests produced a statistically significant result. This result means that when factoring in the additive difference test to the ALREADY statistically significant result from the multiplicative difference test, there is no longer a statistically significant result, implying that the advantage comes in smaller additive distance differences. This seems normal, as in smaller additive distance differences, the team with the advantage would be closer to the location of the match. Looking at the High Seeded and 13-16 results, if the multiplicative distance difference factor was present, but the additive was not, 16/24 High Seeded Teams covered, and 6/7 13-16 Seeded Teams covered. These results are not statistically significant, but suggest that teams with a multiplicative distance advantage that are in close proximity to the city will be where the advantage, if any, lies.

7 Further Study

The obvious extension of this is to find the spread and scoring data for more games in the Round of 64 in years before 2011, and the games missing from the current year's data. More data points will be able to tell us about changes over time and see if the trends currently seen are true or just a byproduct of a small sample size.

8 Sources

- 1. CBS Sports
- 2. FOX Sports
- 3. Sports-Reference
- 4. SB Nation
- 5. Bleacher Report